MA385: Tutorial Sheet 1

These exercises are for Tutorial 1 (Week 3). You do not have to submit solutions to these questions. However, similar questions may appear on the assignment, class test, or final exam.

Q1. Let $f(x) = e^x - 2x - 2$.

(a) Show that there is a solution to the problem: *find* $\tau \in [0, 2]$ *such that* $f(\tau) = 0$.

(b) Taking $x_0 = 0$ and $x_1 = 2$, use 6 steps of the bisection method to estimate τ (you may use a computer program to do this, if you like).

(c) Give an upper bound for the error $|\tau - x_6|$ using Theorem 1.2.1.

(d) What is the smallest k needed to ensure that $|x_k - \tau| \leqslant 10^{-4}$?

Q2. We wish to estimate $\tau = \sqrt[3]{4}$ numerically by solving $f(x) = 0$ in $[a, b]$ for some suitably chosen f, a and b.

(i) Suggest suitable choices of f, a, and b for this problem.

(ii) Show that f has a zero in $[a, b]$.

(iii) Use 6 steps of the bisection method to estimate $\sqrt[3]{4}$. Again, it is OK to write code to do this for you.

(iv) Use Theorem 1.2.1 to give an upper bound for the error $|\tau - x_6|$.

Q3. In lectures, the Secant iteration is defined as

$$x_{k+1} = x_k - f(x_k)\frac{x_k - x_{k-1}}{f(x_k) - f(x_{k-1})} \text{ for } k = 1, 2, 3, \ldots.$$

Show that it can also be written as a *weighted average*

$$x_{k+1} = (1 - \sigma_k)x_k + \sigma_k x_{k-1},$$

and give a formula for σ_k. Derive a formula for σ_k in terms of $f(x_{k-1})$ and $f(x_k)$. Can you give a physical interpretation to σ_k and $1 - \sigma_k$ that might justify the claim that the method should be more efficient than bisection?

Q4. Suppose we have a function f with the property that $f(0) = 1$, $f(1) = -1$ and $f(2) = -1$.

(a) f is a quadratic polynomial: find it. Show that there is a solution to: *find* $\tau \in [0, 2]$ *such that* $f(\tau) = 0$.

(b) Show that the Secant Method will fail when we take $x_0 = 0$ and $x_1 = 2$.

(c) The Secant Method can be *very* sensitive to the choice of x_0 and x_1. By computing a few iterations with $x_0 = 2$ and $x_1 = 0$ show that it appears to converge. (You can write some code to do this, if you like).

Q5. We claimed in class that, if the Secant Method computes the iterates $\{x_0, x_1, x_2 \ldots\}$ then

$$|\tau - x_{k+1}| \leqslant C_k|\tau - x_k||\tau - x_{k-1}|.$$

That is, *the error is bounded by the product of the error in the previous two iterates*. As a step towards this justifying this, show that

$$\tau - x_{k+1} = -(\tau - x_k)(\tau - x_{k-1})\frac{f[x_{k-1}, x_k, \tau]}{f[x_{k-1}, x_k]},$$

where

$$f[a, b] := \frac{f(b) - f(a)}{b - a} \qquad \text{and} \qquad f[a, b, c] := \frac{f[b, c] - f[a, b]}{c - a}.$$